

Geophysical Research Letters[®]



RESEARCH LETTER

10.1029/2026GL125529

Key Points:

- The transition to double-cell mock Walker circulations with warming is associated with the emergence of periodic convection
- Convective and stratiform mode dominate the warm pool alternately in a warmer climate, resulting in the lower and upper cell respectively
- The long-lived stratiform mode after the termination of the deep convection is the key to the double-cell circulation

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

H. Quan,
hengquan@princeton.edu

Citation:

Quan, H., Zhang, Y., Dagan, G., & Fueglistaler, S. (2026). Transition to double-cell mock Walker circulations with surface warming explained by periodic convection. *Geophysical Research Letters*, 53, e2026GL125529. <https://doi.org/10.1029/2026GL125529>

Received 8 JUL 2026

Accepted 16 SEP 2026

Transition to Double-Cell Mock Walker Circulations With Surface Warming Explained by Periodic Convection

Heng Quan^{1,2} , Yi Zhang³ , Guy Dagan⁴ , and Stephan Fueglistaler^{1,2} 

¹Department of Geosciences, Princeton University, Princeton, NJ, USA, ²Program in Atmospheric and Oceanic Sciences, Princeton University, Princeton, NJ, USA, ³Courant Institute of Mathematical Sciences, New York University, New York, NY, USA, ⁴Institute of Earth Sciences, The Hebrew University of Jerusalem, Jerusalem, Israel

Abstract Idealized mock Walker simulations are widely used to study the interactions between overturning circulation and convection in the tropics. Previous studies documented a transition from a single-cell to a double-cell mock Walker circulation when the average sea surface temperature exceeds 300 K. Here, we ascribe the transition to the emergence of periodic convection with warming due to stronger convectively coupled waves. In cold simulations, the warm pool is dominated by steady deep convection, which results in a single overturning cell. In hot simulations, the warm pool is alternately dominated by deep convection and a stratiform mode, resulting in lower and upper cells respectively. This study suggests that the idealized mock Walker circulation in a warmer climate features complex structural changes in addition to a weakening in strength, and highlights the profound impacts of convection on the structure of the overturning circulation.

Plain Language Summary The Walker circulation features air rising over the warm western Pacific, flowing eastward high up, sinking over the cooler eastern Pacific, and returning near the surface. Previous studies ran idealized atmospheric model simulations, and found that the single-cell Walker circulation splits into two vertically stacked overturning cells at a critical surface temperature close to today's tropical Pacific average. Here, we trace the transition to how the rising air behaves over the warmest, rainiest region. In a cold simulation, this region is always dominated by air ascending from the surface to the top, which drives a single overturning cell. In a hot simulation, this region is alternately dominated by air ascending from the surface to the top, which drives the lower overturning cell; and air ascending only in the upper level, which drives the upper overturning cell. Together, averaged over time, they form the double-cell structure. Our results suggest that as the climate warms, the tropical atmospheric circulation in idealized simulations may not simply weaken but could change shape, implying that localized vertical motion may affect the structure of basin-scale overturning circulation.

1. Introduction

The tropical atmospheric motion has multiple scales ranging from planetary-scale overturning circulation to kilometer-scale convective processes. Multi-scale interactions are important for the tropical climate because they affect circulation patterns, cloud distributions, and their responses to global warming (Bony et al., 2015). The mock Walker simulation (Kuang, 2012; Raymond, 1994; Wofsy & Kuang, 2012) is an idealized tool to study the coupling between overturning circulation and convection. It is a cloud-resolving simulation on a rectangular domain with sea surface temperature (SST) uniform in the short dimension and varying in the long dimension, mimicking the zonal SST gradient in the equatorial Pacific and deriving a Walker-like overturning circulation. The length of the channel ($\mathcal{O}(10000\text{ km})$) is large enough to resolve tropical-scale circulations, and the resolution ($\mathcal{O}(1\text{ km})$) is fine enough to partially resolve small-scale convection. The mock Walker simulation has been used to study tropical circulation (Grabowski et al., 2000; Kuang, 2012; Quan, Zhang, & Fueglistaler, 2025; Yano et al., 2002), the responses of tropical precipitation to global warming (Dagan, Seeley, & Steiger, 2023; Quan et al., 2026; Sokol et al., 2026), and tropical climate feedback (Quan, Zhang, Wang, & Fueglistaler, 2025; Wing et al., 2024) and forcing (Dagan, Yeheskel, & Williams, 2023; Sreelekshmi et al., 2026).

Besides the expected weakening of the mean overturning circulation with warming (Held & Soden, 2006; Vecchi et al., 2006), previous studies using mock Walker simulations documented a transition from a single-cell to a double-cell circulation structure with two vertically stacked overturning cells when the average SST becomes

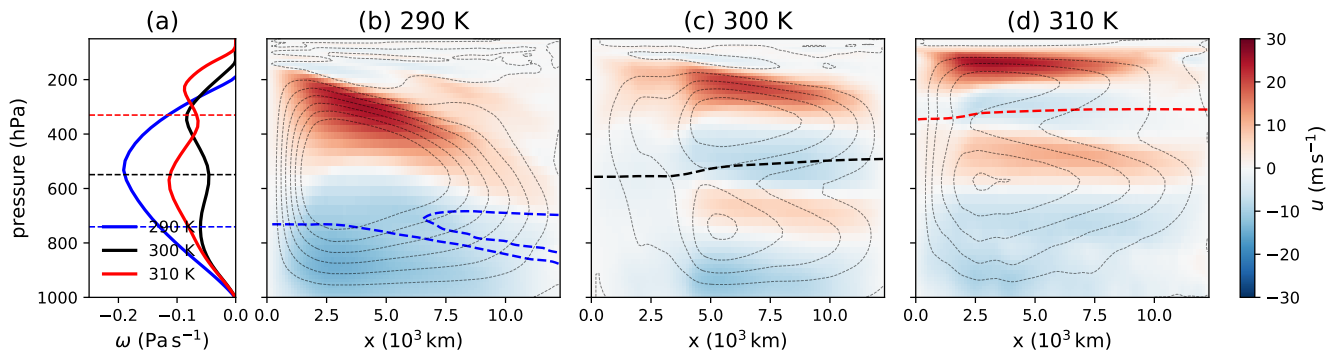


Figure 1. (a) The vertical velocity ω for 3-D SAM mock Walker simulations with mean SSTs of 290 K (blue), 300 K (black) and 310 K (red) averaged over time and over the warm pool (defined as regions with the top 20% precipitation). (b–d) The time-averaged zonal wind u for 3-D SAM mock Walker simulations (with the warm pool in the left of the domain). The gray contours in the background show the streamfunction $\Psi(x, p) = \int_0^x \omega(x', p) dx'$ with an interval of $50,000 \text{ kg m}^{-1} \text{ s}^{-1}$. Solid contours indicate counterclockwise flow and dashed contours indicate clockwise flow. The colored dashed lines in all panels are 0°C isotherms.

higher (Grabowski et al., 2000; Yano et al., 2002; N. J. Lutsko & Cronin, 2024; Quan, Zhang, & Fueglistaler, 2025) (also see our Figure 1). This transition affects the distributions of convection and clouds, but the reason for the transition remains puzzling.

A recent study by N. J. Lutsko and Cronin (2024) related the transition to double-cell circulations to the vertical profile of stability over the cold pool. They demonstrated that a mid-tropospheric maximum stability emerges with SST warming for a moist adiabat (their Text S3), which is consistent with a mid-tropospheric minimum descending velocity over the cold pool, as well as a double-cell circulation according to continuity. However, this is not a causal argument. Only the lapse rate over the warm pool is strongly constrained toward a moist adiabatic lapse rate, whereas the lapse rate over the cold pool is not. The free-tropospheric temperature over the cold pool is controlled by the warm pool and is detached from the surface (Zhang & Fueglistaler, 2020). Therefore, although intriguing, the mid-tropospheric maximum stability emphasized by N. J. Lutsko and Cronin (2024) cannot fully explain the transition to double-cell mock Walker circulation with warming.

Here, we explain the transition to double-cell mock Walker circulations with warming through the emergence of periodic convection over the warm pool (Dagan, Seeley, & Steiger, 2023; Quan et al., 2026; Seeley & Wordsworth, 2021; Song et al., 2024; Spaulding-Astudillo & Mitchell, 2024). A recent study found that precipitation over the warm pool transitions from a steady state to an intense periodic oscillation state as the average SST warms (Quan et al., 2026), which is ascribed to the intensification of large-scale convectively coupled waves. We show that the warm pool is alternately dominated by deep convection (with strong surface precipitation) and a stratiform mode (with little surface precipitation) in hot simulations. Deep convection in the entire troposphere results in the lower overturning cell, while the stratiform mode confined to the upper-troposphere results in the upper overturning cell. By contrast, the warm pool is always dominated by steady deep convection in cold simulations, resulting in a single overturning cell. Our results advance the understanding of tropical circulation structure in a warmer climate, and highlight the importance of the two-way interaction between convection and large-scale dynamics.

The paper is organized as follows. Section 2 describes the mock Walker simulations used in this study. Section 3 relates the transition to double-cell mock Walker circulations to the emergence of periodic convection over the warm pool, and demonstrates that periodic convection is necessary for the double-cell. Finally, Section 4 summarizes the conclusions and discusses implications.

2. Models and Simulations

We use the System for Atmospheric Modeling (SAM) (Khairoutdinov & Randall, 2003) version 6.11.5 cloud-resolving model (CRM) to conduct mock Walker simulations (Kuang, 2012) with prescribed SSTs. Full details of the model are provided in Text S1 in Supporting Information S1. In the main text, we focus on 3-D mock Walker simulations with a domain size of $12288 \text{ km} \times 128 \text{ km}$. We use solid wall boundary conditions at the two edges in the long dimension and periodic boundary conditions in the short dimension. Following Quan, Zhang, and Fueglistaler (2025), the prescribed SSTs linearly decrease by 8 K from the left boundary ($x = 0$) to the right boundary

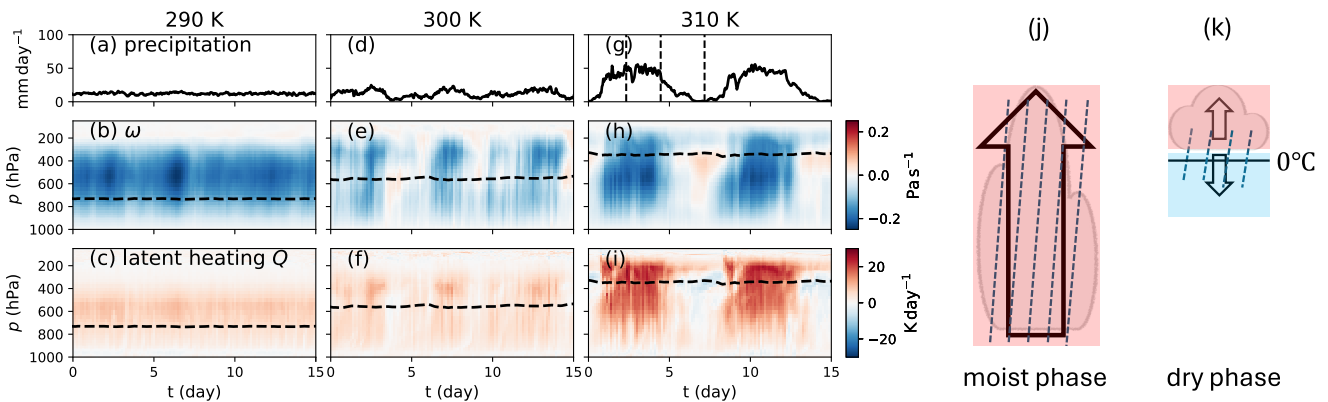


Figure 2. (a, d, g) The 15-day time series of hourly precipitation averaged over the warm pool for 3-D SAM mock Walker simulations with mean SSTs of 290 K, 300 and 310 K. The three time slices marked in (g) correspond to three snapshots analyzed in Figure 3. (b, e, h) The corresponding vertical velocity (ω) averaged over the warm pool. (c, f, i) The corresponding latent heating rate (Q) averaged over the warm pool. In panels (b, c, e, f, h and i) the dashed lines mark the 0°C isotherms. (j, k) Schematics of the moist phases (with high surface precipitation) and the dry phases (with little surface precipitation) over the warm pool of the 310 K simulation. Arrows represent vertical motion, red shadings represent net latent heating and blue shadings represent net latent cooling.

($x = 12288$ km), mimicking the east-west SST gradient across the equatorial Pacific and causing an overturning circulation. The domain average SST ranges from 290 to 310 K with an increment of $+5$ K between simulations. All 3-D simulations use interactive radiation, with the incoming solar radiation fixed at 650.83 W m^{-2} (no diurnal cycle). All simulations are run for 140 days and reach equilibrium after approximately 50 days. All our results are based on the last 40 days of hourly model output and are averaged along the short dimension. We define the warm pool as the regions with the top 20% time-mean precipitation; a different choice (e.g., the top 10% SST) does not affect the results (not shown). Text S1 in Supporting Information S1 details additional simulations with different setups, and Table S1 in Supporting Information S1 provides a complete list of all simulations.

3. Results

3.1. Transition to Double-Cell Mock Walker Circulations With Warming

Figures 1b–1d reproduces the transition to double-cell mock Walker circulations with surface warming shown in N. J. Lutsko and Cronin (2024). The time-averaged zonal wind u features a single overturning cell in the 290 K mock Walker simulation, whereas the 300 and 310 K simulations feature two vertically stacked overturning cells. The SST threshold for the transition, 300 K, is close to the present average SST in the equatorial Pacific. The double-cell structure expands upwards with SST warming due to the deepening of the troposphere. The upper cell lies between the 600 hPa level and the 100 hPa level in the 300 K simulation, and moves up to above the 400 hPa level in the 310 K simulation.

The double-cell circulation has a double-peak profile of the vertical velocity (ω) according to the continuity relation. Figure 1a shows ω averaged over the warm pool in three simulations. $\omega(p)$ has a single peak corresponding to the single overturning cell in the 290 K simulation, while it has two peaks corresponding to the lower and upper cells in the 300 and 310 K simulations. The local minimum in ω between the two peaks is very close to the melting level (i.e., the 0°C isotherm), which, as we will show later, is not a coincidence.

The time-averaged circulation structures above are robust to simulation set-ups. Figure S1 in Supporting Information S1 shows that the transition to double-cell mock Walker circulations with surface warming occurs in 2-D or 3-D domains; with single- or double-moment microphysics; and with interactive or fixed radiation. In all cases the SST threshold for the transition is around 300 K, and the local minimum ω between two peaks, when present, is close to the melting level.

3.2. Double-Cell Circulation Explained by Periodic Convection

The double-peak $\omega(p)$ profile in Figure 1a co-occurs with a transition from steady convection to periodic convection as the SST warms. Figure 2 shows the time evolution of precipitation, the vertical velocity (ω) and the latent heating rate (Q) averaged over the warm pool in a selected 15-day period. The 290 K simulation

(Figures 2a–2c) features steady deep convection with ascending motion and net latent heating throughout the free troposphere. The 300 K simulation (Figure 2d–2f) is in an intermediate state with weak oscillatory behavior. The 310 K simulation (Figures 2g–2i) features intense periodic oscillation in precipitation, ω and Q . The periodicity of convection increases with increasing SST (Figure S2 in Supporting Information S1), which co-occurs with the transition to double-cell circulation. This transition from steady convection to periodic convection with warming is robust to simulation set-ups (Figure S3 in Supporting Information S1) and can be explained by the intensification of large-scale convectively coupled waves (Quan et al., 2026): Large-scale convectively coupled waves become unstable (i.e., the growth rate is positive) when the SST approaches 305 K, close to the SST threshold (around 300 K) for the transition to double-cell circulation.

Moist and dry phases of the periodic convection in the warm pool of the 310 K simulation have distinct vertical structures. During moist phases (e.g., day 3) with large surface precipitation, there is ascending motion and net latent heating throughout the free troposphere. As summarized in Figure 2j, moist phases are dominated by deep convection. During dry phases (e.g., day 7) with little surface precipitation, weaker ascending motion and net latent heating are confined to the upper-troposphere above the 300 hPa level, while subsidence and net latent cooling dominate the mid-troposphere. As summarized in Figure 2k, dry phases are dominated by the stratiform mode (Kuang, 2008; Mapes, 2000), where the condensates aloft hardly fall to the surface due to re-evaporation in the mid-troposphere. The vertical profiles of ω averaged over moist phases and dry phases in Figure S4 in Supporting Information S1 are consistent with the schematics in Figures 2j and 2k. They are unlikely artifacts of the microphysical parameterization because they persist when replacing the default single-moment microphysics scheme (Khairoutdinov & Randall, 2003) with a double-moment scheme (Morrison et al., 2005) (Figure S5 in Supporting Information S1).

The periodic convection explains the time-averaged double-peak ω profile in Figure 1a and the double-cell circulation. In the 310 K simulation, deep convection and the stratiform mode dominate the warm pool alternately (Figures 2h and 2i). Deep convection results in the lower peak of the time-averaged ω in Figure 1a and the lower overturning cell (“lower cell” in this paper refers to the deep circulation with the lower peak in time-averaged ω), while the upper-tropospheric ascending motion of the stratiform mode results in the upper peak of the time-averaged ω and the upper overturning cell. The subsidence near the melting level of the stratiform mode results in the local minimum time-averaged ascending velocity near the melting level. By contrast, in the 290 K simulation, the warm pool is always dominated by steady deep convection (Figures 2b and 2c), which results in a single peak of the time-averaged ω in Figure 1a and a single overturning cell.

The stratiform mode in our simulations resembles the stratiform regions in mesoscale convective systems (MCSs). As illustrated in Figure 3a, the upper-level outflow of the deep convection is directed rearward relative to a moving MCS, creating a front-to-rear slantwise ascending motion (Pandya & Durran, 1996) and anvil clouds due to condensation. The region below the anvil cloud base features a rear-to-front inflow as a gravity wave response to latent heating in the convective region (Pandya & Durran, 1996), and this inflow is forced to descend due to the latent cooling of falling precipitation particles (Houze Jr, 2004). The latent cooling peaks around the melting level where melting, evaporation and sublimation act together (Fu & O’Neill, 2024), but the subsidence and net latent cooling start above the melting level due to sublimation (Braun & Houze Jr, 1997; Gamache & Houze Jr, 1982). A typical MCS has a $\mathcal{O}(100\text{ km})$ horizontal scale and a $\mathcal{O}(10\text{ hr})$ lifetime (Chen & Houze Jr, 1997), but the structure in Figure 3a also occurs in large-scale convectively coupled waves that can span thousands of kilometers and last for several days (Kiladis et al., 2009). The ω and Q averaged over the warm pool in the dry phases of our 310 K mock Walker simulation (Figures 2h, 2i, and 2k) show a similar dipole structure to the stratiform region in an MCS, with subsidence and net latent cooling starting above the melting level and peaking around the melting level. While the node (where $\omega \approx 0$, $Q \approx 0$) of the stratiform mode is close to the middle of the troposphere in the present climate (Kuang, 2008; Mapes, 2000), it shifts upward in the 310 K simulation due to a higher melting level (as shown in Figure 2k), so the dipole structure is confined to the upper level.

The snapshots at days 2.33, 4.46 and 7.17 in Figures 3b–3m confirm that the generation of the stratiform mode in the 310 K mock Walker simulation is similar to the generation of the stratiform region in MCSs. The three time slices are marked in Figure 2g and are comparable to the growing, mature and dissipating stages in the life cycle of an MCS (Robert, 1982).

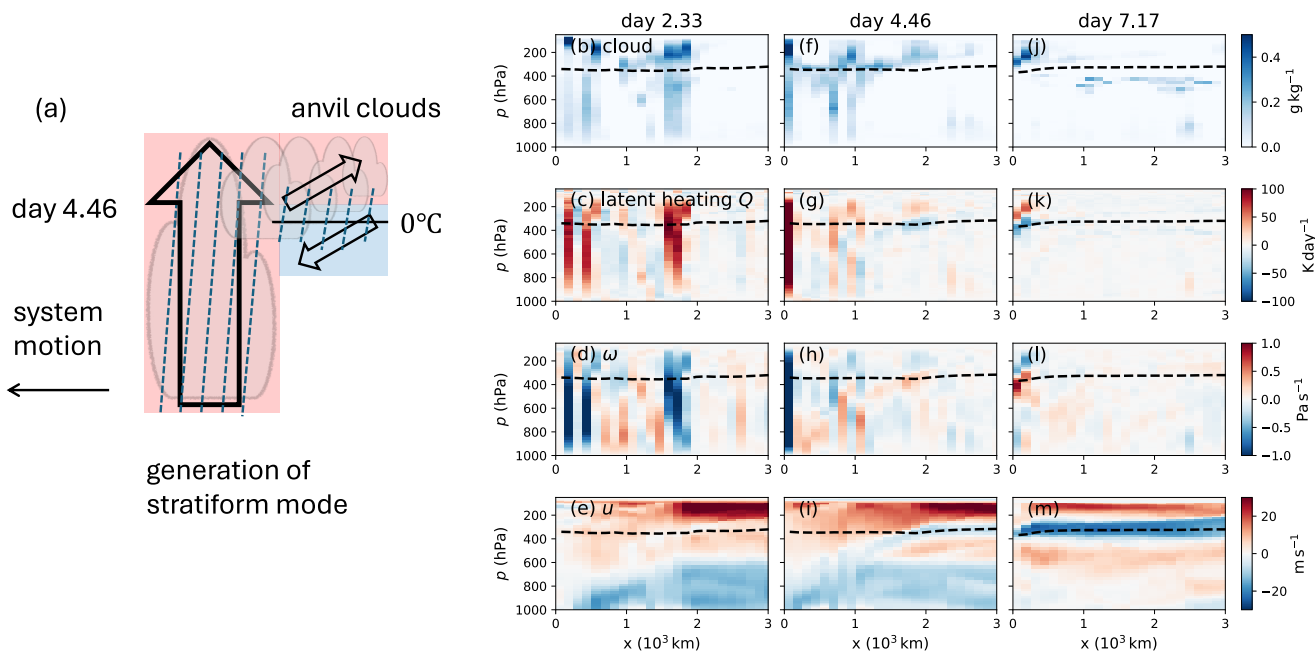


Figure 3. (a) A schematic showing the generation of the stratiform region in a mesoscale convective system. (b)–(e) The x - p distribution of the non-precipitating cloud condensates, the latent heating rate (Q), the vertical velocity (ω) and the zonal wind (u) at day 2.33 over the warm pool of the 310 K mock Walker simulation. Note that x ranges from 0 to 3,000 km while the length of the domain is 12,288 km. (f)–(i) Same as (b)–(e) but at day 4.46. (j)–(m) Same as (b)–(e) but at day 7.17. The dashed lines in (b)–(m) mark the 0°C isotherms.

1. At day 2.33 (Figures 3b–3e), there are two deep convective cells at $x \approx 500$ km and $x \approx 1800$ km featuring deep convective clouds, net latent heating, and ascending motion throughout the entire free troposphere. The zonal wind associated with deep convection in panel (e) shows convergence (toward the solid wall) below 600 hPa and divergence (away from the solid wall) above 600 hPa, corresponding to the lower overturning cell in Figure 1d.
2. At day 4.46 (Figures 3f–3i), there is a clear stratiform region ($1000 \text{ km} < x < 2000 \text{ km}$) behind the deep convective cell (Figure S6E in Quan et al. (2026) indicates that convectively coupled waves propagate toward the warm pool in this simulation.). A front-to-rear slantwise ascending motion creates anvil clouds due to net condensation in the stratiform region, while below the cloud base a rear-to-front inflow is forced to descend due to net latent cooling before reaching the melting level. These structures resemble the generation of the stratiform region in an MCS (Figure 3a). The zonal wind in panel (i) shows both the lower overturning cell (at $x < 1000$ km; “lower cell” in this paper refers to the deep circulation with the lower peak in time-averaged ω) associated with the deep convection and an emerging upper overturning cell (at $x > 1000$ km) confined above 400 hPa associated with the stratiform mode. The vertical shear of the zonal wind (leftward in the lower level, rightward in the upper level) helps to tilt convection and reinforces the generation of the stratiform region.
3. At day 7.17 (Figures 3j–3m), the deep convection terminates, but the stratiform mode remains (at $x < 400$ km) in the upper troposphere. The upper overturning cell above 400 hPa associated with the stratiform mode in panel (m) contributes to the time-averaged upper overturning cell in Figure 1d.

A video version of Figures 3b–3m for mock Walker simulations with different SSTs further supports that the periodic convection in the warm pool causes the double-cell overturning circulation (Movie S1). In the 290 K simulation, stratiform regions occasionally emerge behind the deep convection, but there is no clear temporal separation between deep convection and the stratiform mode. Steady deep convection always dominates over the stratiform mode in the warm pool (this is also evident in Figures 2b and 2c), so the upper overturning cell associated with the stratiform mode never emerges. By contrast, in the 310 K simulation, there is clear temporal separation between deep convection and the stratiform mode due to the periodic oscillation. In moist phases, deep convection dominates in the warm pool and creates the lower overturning cell (Figure 3e); in dry phases, the stratiform mode dominates and creates the upper overturning cell (Figure 3m). The stratiform mode alone can

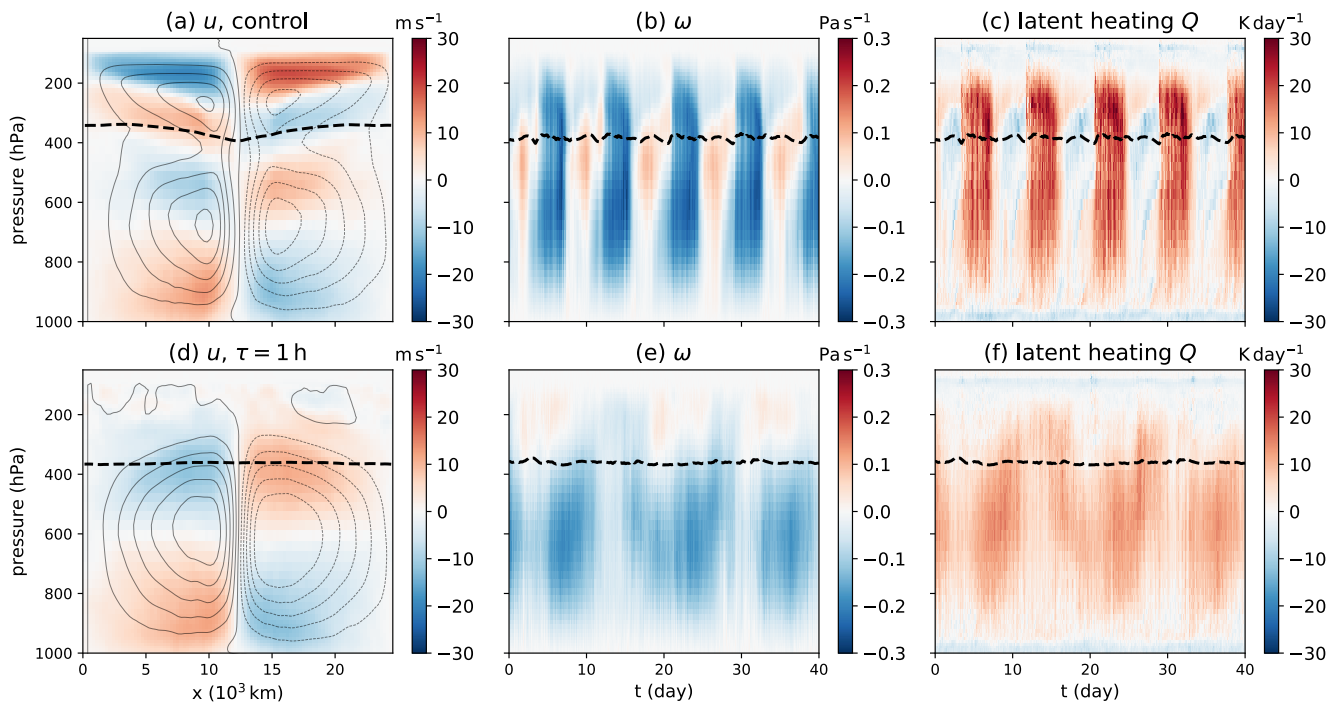


Figure 4. (a) The time-averaged zonal wind (u) for a 2-D SAM mock Walker control simulation with the warm pool at the center (details in text). The gray contours in the background show the streamfunction $\Psi(x, p) = \int_0^x \omega(x', p) dx'$ with an interval of $50,000 \text{ kg m}^{-1} \text{ s}^{-1}$. Solid contours indicate counterclockwise flow and dashed contours indicate clockwise flow. (b, c) The 40-day evolution of vertical velocity (ω) and latent heating rate (Q) averaged over the warm pool of the control simulation. (d, e, f) Same as (a, b, c), but for a mechanism-denial simulation that is identical to the control simulation except for an additional moisture source in the mid-troposphere (details in text). The black dashed lines mark the 0°C isotherms.

persist for 2–3 days during dry phases until deep convection recurs, and this persistence is the key to the time-averaged double-cell circulation (Figure S4 in Supporting Information S1).

Before moving on, we briefly comment on the role of ice and the melting level. Because of strong localized latent cooling, the melting level emerges as the level with a local minimum ω in Figure 1a. However, it is not a necessary ingredient for the double-cell circulation. Figure S5 in Supporting Information S1 demonstrates that the double-cell circulation still exists in the 310 K simulation without ice (i.e., water only has a vapor phase and a liquid phase). The theory for the transition to periodic convection with warming (Quan et al., 2026) invokes large-scale convectively coupled waves and is not dependent on the presence of ice, so the ice-free 310 K simulation still features periodic convection in the warm pool (Figure S5 in Supporting Information S1), which causes the double-cell circulation.

3.3. Mechanism-Denial Simulations Confirm the Necessity of Periodic Convection for the Double-Cell Circulation

We design a mechanism-denial experiment to demonstrate the necessity of periodic convection for the double-cell circulation. In this case, our control simulation is a 2-D (x - z) mock Walker simulation with a sinusoidal SST profile. Following N. J. Lutsko and Cronin (2024), the average SST is 310 K, and the difference between the warmest SST at the center and the coldest SST at the edges is 5 K. The length of the domain is 24,576 km, and we use a periodic boundary condition. Other settings are identical to Section 2. Consistent with the results above, the control simulation features a double-cell overturning circulation (Figure 4a) due to periodic convection in the warm pool (Figures 4b and 4c), which also shows that deep convection and the stratiform mode dominate the warm pool alternately.

The mechanism-denial simulation features steady deep convection in the warm pool. It is identical to the control simulation, except for an artificial moisture source in the model's moisture budget equation:

$$\frac{\partial q}{\partial t} = \dots + \max\left\{\frac{0.75q^* - q}{\tau}, 0\right\}, \quad (1)$$

where q is the specific humidity, q^* is the saturation specific humidity, and $\tau = 1$ h is a moistening timescale. The artificial moisture source is added only when $q/q^* < 0.75$ and $263.15 \text{ K} < T < 283.15 \text{ K}$ to keep the mid-tropospheric relative humidity (RH) above 75%. Intuitively, the artificial moisture source enhances deep convection by moistening the mid-troposphere and reducing the evaporative cooling from entrainment (Ahmed et al., 2020; Kuang, 2008; Weber et al., 2021), so we expect that deep convection in the warm pool remains steady and does not decay in the mechanism-denial simulation. In the context of Quan et al. (2026), the termination of deep convection in the periodic oscillation is related to a positive feedback: An anomalous subsidence perturbation in the mid-troposphere results in a dry anomaly due to vertical moisture advection, which causes an anomalous evaporative cooling that amplifies the subsidence perturbation. The artificial moisture source in the mechanism-denial simulation breaks this positive feedback; hence, it should maintain steady deep convection in the warm pool. Indeed, Figures 4e and 4f shows steady deep convection in the mechanism-denial simulation. Although a weaker oscillatory behavior still exists, the warm pool is always dominated by deep convection, while the stratiform-only mode never emerges.

With steady deep convection in the warm pool, the mechanism-denial simulation has a single-cell overturning circulation instead of a double-cell structure (Figure 4d). The lower peak of the warm-pool ω around 600 hPa and the lower cell associated with deep convection are similar to those in the control simulation, while the upper peak of the warm-pool ω around 300 hPa and the upper cell associated with the stratiform mode disappear (Figure S6 in Supporting Information S1). Therefore, the mechanism-denial simulation supports that periodic convection is necessary for the double-cell circulation. While aiming to remove the periodic convection, the mechanism-denial simulation inevitably increases the mean-state mid-tropospheric moisture. It is unclear whether the mean-state mid-tropospheric moisture can directly affect the circulation structure regardless of the periodicity in convection, and this warrants further research.

4. Conclusions and Discussions

We study the transition from single-cell to double-cell mock Walker circulations with SST warming, and ascribe it to the emergence of periodic convection in the warm pool discussed in our preceding paper (Quan et al., 2026). In a cold simulation, the warm pool is always dominated by steady deep convection, which results in a single overturning cell. In a hot simulation, the warm pool is alternately dominated by deep convection (in moist phases) and the stratiform mode (in dry phases). They create the lower and upper cells, respectively, resulting in a time-averaged double-cell circulation. We show that the generation of the stratiform mode is similar to the generation of the stratiform region in a moving MCS, and the persistence of the stratiform mode without deep convection is the key to the double-cell circulation. Our mechanism-denial simulations with periodic convection disabled further confirm that periodic convection is necessary for the double-cell circulation.

Our results help to better understand an earlier study by N. J. Lutsko and Cronin (2024), which related the transition to double-cell mock Walker circulations to an emerging mid-tropospheric minimum in radiative-subsidence velocity over the cold pool and an associated maximum static stability. Periodic convection generates a double-peak $\omega(p)$ profile (ascending) in a warmer climate (Figure 1a) over the warm pool, which results in the double-peak $\omega(p)$ profile (subsiding) over the cold pool (and a minimum between the two peaks) found by N. J. Lutsko and Cronin (2024), as required by mass conservation. The double-peak $\omega(p)$ profile (subsiding) could shape the profile of static stability via vertical temperature advection.

One limitation of this study is that we only use a single cloud-resolving model. We also analyzed the output of mock Walker simulations based on other cloud-resolving models in RCEMIP-II (Wing et al., 2024), but most of them have an SST contrast of 1.25 K between the warm pool and the cold pool, which is not large enough to generate a clear overturning circulation comparable to our Figures 1b–1d (results not shown). Therefore, mock Walker simulations based on other models and with larger (> 5 K) SST contrasts can help verify the theories in this study.

Meanwhile, caution should be taken when extrapolating the results of this study, based on idealized simulations, to the real atmosphere. While the SST threshold (near 300 K) for the transition to double-cell mock Walker

circulations is close to the present-day average SST in the equatorial Pacific, there is no double-cell circulation in the reanalysis data (N. J. Lutsko & Cronin, 2024). The Walker circulation features a faster weakening in the upper-troposphere than near the surface under global warming in CMIP6 coupled global climate models (GCMs), but this upper-lower tropospheric decoupling occurs without an obvious transition to the double-cell structure (Wu et al., 2024). It is possible that the double-cell structure cannot emerge with warming in the real world, but it is also possible that GCMs with a resolution of around 100 km cannot accurately show the transition to periodic convection, which is necessary for the double-cell structure. Future work may run global storm resolving models with km-scale resolution and check whether a double-cell Walker circulation emerges together with periodic convection in the warm pool in a warmer climate.

Nevertheless, this study suggests that the Walker circulation in a warmer climate might feature complex structural changes in addition to a weakening of strength, and highlights the importance of the two-way interaction between convection and large-scale dynamics.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The source code for SAM with the artificial moisture source (Equation 1) can be found at <https://zenodo.org/records/12646430> (N. Lutsko, 2024). Post-processed model output and plotting scripts are available at <https://doi.org/10.34770/b7fw-nd77> (Quan, 2026).

Acknowledgments

The authors acknowledge two anonymous reviewers for their helpful comments. The authors gratefully acknowledge Marat F. Khairoutdinov for creating and maintaining SAM and Nicholas Lutsko for developing the method that adds the artificial moisture source to SAM. This report was prepared by Heng Quan under Award NA23OAR4320198-T1-01 from the National Oceanic and Atmospheric Administration, U.S. Department of Commerce. The statements, findings, conclusions, and recommendations are those of the author(s) and do not necessarily reflect the views of the National Oceanic and Atmospheric Administration, or the U.S. Department of Commerce.

References

- Ahmed, F., Adames, Á. F., & Neelin, J. D. (2020). Deep convective adjustment of temperature and moisture. *Journal of the Atmospheric Sciences*, 77(6), 2163–2186. <https://doi.org/10.1175/jas-d-19-0227.1>
- Bony, S., Stevens, B., Frierson, D. M., Jakob, C., Kageyama, M., Pincus, R., et al. (2015). Clouds, circulation and climate sensitivity. *Nature Geoscience*, 8(4), 261–268. <https://doi.org/10.1038/ngeo2398>
- Braun, S. A., & Houze Jr, R. A. (1997). The evolution of the 10–11 June 1985 PRE-STORM squall line: Initiation, development of rear inflow, and dissipation. *Monthly Weather Review*, 125(4), 478–504. [https://doi.org/10.1175/1520-0493\(1997\)125<0478:teotjp>2.0.co;2](https://doi.org/10.1175/1520-0493(1997)125<0478:teotjp>2.0.co;2)
- Chen, S. S., & Houze Jr, R. A. (1997). Diurnal variation and life-cycle of deep convective systems over the tropical Pacific warm pool. *Quarterly Journal of the Royal Meteorological Society*, 123(538), 357–388. <https://doi.org/10.1002/qj.49712353806>
- Dagan, G., Seeley, J. T., & Steiger, N. (2023). Convection and convective-organization in hothouse climates. *Journal of Advances in Modeling Earth Systems*, 15(11), e2023MS003765. <https://doi.org/10.1029/2023ms003765>
- Dagan, G., Yehekel, N., & Williams, A. I. (2023). Radiative forcing from aerosol–cloud interactions enhanced by large-scale circulation adjustments. *Nature Geoscience*, 16(12), 1092–1098. <https://doi.org/10.1038/s41561-023-01319-8>
- Fu, H., & O'Neill, M. E. (2024). The small-amplitude dynamics of spontaneous tropical cyclogenesis. Part II: Linear stability analysis. *Journal of the Atmospheric Sciences*, 81(12), 2069–2091. <https://doi.org/10.1175/jas-d-24-0032.1>
- Gamache, J. F., & Houze Jr, R. A. (1982). Mesoscale air motions associated with a tropical squall line. *Monthly Weather Review*, 110(2), 118–135. [https://doi.org/10.1175/1520-0493\(1982\)110<0118:mamawa>2.0.co;2](https://doi.org/10.1175/1520-0493(1982)110<0118:mamawa>2.0.co;2)
- Grabowski, W. W., Yano, J.-I., & Moncrieff, M. W. (2000). Cloud resolving modeling of tropical circulations driven by large-scale SST gradients. *Journal of the Atmospheric Sciences*, 57(13), 2022–2040. [https://doi.org/10.1175/1520-0469\(2000\)057<2022:crmote>2.0.co;2](https://doi.org/10.1175/1520-0469(2000)057<2022:crmote>2.0.co;2)
- Held, I. M., & Soden, B. J. (2006). Robust responses of the hydrological cycle to global warming. *Journal of Climate*, 19(21), 5686–5699. <https://doi.org/10.1175/jcli3990.1>
- Houze Jr, R. A. (2004). Mesoscale convective systems. *Reviews of Geophysics*, 42(4). <https://doi.org/10.1029/2004rg000150>
- Khairoutdinov, M. F., & Randall, D. A. (2003). Cloud resolving modeling of the ARM summer 1997 IOP: Model formulation, results, uncertainties, and sensitivities. *Journal of the Atmospheric Sciences*, 60(4), 607–625. [https://doi.org/10.1175/1520-0469\(2003\)060<0607:crmota>2.0.co;2](https://doi.org/10.1175/1520-0469(2003)060<0607:crmota>2.0.co;2)
- Kiladis, G. N., Wheeler, M. C., Haertel, P. T., Straub, K. H., & Roundy, P. E. (2009). Convectively coupled equatorial waves. *Reviews of Geophysics*, 47(2). <https://doi.org/10.1029/2008rg000266>
- Kuang, Z. (2008). A moisture-stratiform instability for convectively coupled waves. *Journal of the Atmospheric Sciences*, 65(3), 834–854. <https://doi.org/10.1175/2007jas2444.1>
- Kuang, Z. (2012). Weakly forced mock walker cells. *Journal of the Atmospheric Sciences*, 69(9), 2759–2786. <https://doi.org/10.1175/jas-d-11-0307.1>
- Lutsko, N. (2024). nicklutsko/mock-walker-double-cell: First release (version v1.0.0) [Software]. *Zenodo*. <https://doi.org/10.5281/zenodo.12646430>
- Lutsko, N. J., & Cronin, T. W. (2024). The transition to double-celled circulations in mock-walker simulations. *Geophysical Research Letters*, 51(14), e2024GL108945. <https://doi.org/10.1029/2024gl108945>
- Mapes, B. E. (2000). Convective inhibition, subgrid-scale triggering energy, and stratiform instability in a toy tropical wave model. *Journal of the Atmospheric Sciences*, 57(10), 1515–1535. [https://doi.org/10.1175/1520-0469\(2000\)057<1515:csiste>2.0.co;2](https://doi.org/10.1175/1520-0469(2000)057<1515:csiste>2.0.co;2)
- Morrison, H., Curry, J., & Khvorostyanov, V. (2005). A new double-moment microphysics parameterization for application in cloud and climate models. Part I: Description. *Journal of the Atmospheric Sciences*, 62(6), 1665–1677. <https://doi.org/10.1175/jas3446.1>
- Pandya, R. E., & Durran, D. R. (1996). The influence of convectively generated thermal forcing on the mesoscale circulation around squall lines. *Journal of the Atmospheric Sciences*, 53(20), 2924–2951. [https://doi.org/10.1175/1520-0469\(1996\)053<2924:tiocgt>2.0.co;2](https://doi.org/10.1175/1520-0469(1996)053<2924:tiocgt>2.0.co;2)

- Quan, H. (2026). Data and codes for the paper "transition to double-cell mock Walker circulations with surface warming explained by periodic convection [Dataset]. <https://doi.org/10.34770/b7fw-nd77>
- Quan, H., Zhang, B., Wang, C., & Fueglistaler, S. (2025). The sea surface temperature pattern effect on outgoing longwave radiation: The role of large-scale convective aggregation. *Geophysical Research Letters*, 52(11), e2024GL112756. <https://doi.org/10.1029/2024gl112756>
- Quan, H., Zhang, Y., Dagan, G., & Fueglistaler, S. (2026). Periodic extreme rainfall in a warmer climate due to stronger convectively coupled waves. *Science Advances*, 12(32), eaed1634. <https://doi.org/10.1126/sciadv.aed1634>
- Quan, H., Zhang, Y., & Fueglistaler, S. (2025). Weakening of tropical free-tropospheric temperature gradients with global warming. *Journal of the Atmospheric Sciences*, 82(1), 31–43. <https://doi.org/10.1175/jas-d-24-0140.1>
- Raymond, D. J. (1994). Convective processes and tropical atmospheric circulations. *Quarterly Journal of the Royal Meteorological Society*, 120(520), 1431–1455. <https://doi.org/10.1002/qj.49712052002>
- Robert, A. (1982). Cloud clusters and large-scale vertical motions in the tropics. *Journal of the Meteorological Society of Japan*, 60(1).
- Seeley, J. T., & Wordsworth, R. D. (2021). Episodic deluges in simulated hothouse climates. *Nature*, 599(7883), 74–79. <https://doi.org/10.1038/s41586-021-03919-z>
- Sokol, A. B., Merlis, T. M., & Fueglistaler, S. (2026). No "wet gets wetter" in kilometer-scale mock-walker circulations. *AGU Advances*, 7(1), e2025AV002040. <https://doi.org/10.1029/2025av002040>
- Song, X., Abbot, D. S., & Yang, J. (2024). Critical role of vertical radiative cooling contrast in triggering episodic deluges in small-domain hothouse climates. *Journal of Advances in Modeling Earth Systems*, 16(6), e2023MS003912. <https://doi.org/10.1029/2023ms003912>
- Spaulding-Astudillo, F. E., & Mitchell, J. L. (2024). A simple model for the emergence of relaxation-oscillator convection. *Journal of Advances in Modeling Earth Systems*, 16(11), e2024MS004439. <https://doi.org/10.1029/2024ms004439>
- Sreelekshmi, T., Shpund, J., Habib, N., & Dagan, G. (2026). A hierarchical modeling study of absorbing aerosol impacts on precipitation characteristics and extremes. *Journal of Advances in Modeling Earth Systems*, 18(6), e2025MS005400. <https://doi.org/10.1029/2025ms005400>
- Vecchi, G. A., Soden, B. J., Wittenberg, A. T., Held, I. M., Leetmaa, A., & Harrison, M. J. (2006). Weakening of tropical pacific atmospheric circulation due to anthropogenic forcing. *Nature*, 441(7089), 73–76. <https://doi.org/10.1038/nature04744>
- Weber, N. J., Kim, D., & Mass, C. F. (2021). Convection–Kelvin wave coupling in a global convection-permitting model. *Journal of the Atmospheric Sciences*, 78(4), 1039–1055. <https://doi.org/10.1175/jas-d-20-0243.1>
- Wing, A. A., Silvers, L. G., & Reed, K. A. (2024). Rcmip-ii: Mock-walker simulations as phase ii of the radiative–convective equilibrium model intercomparison project. *Geoscientific Model Development*, 17(16), 6195–6225. <https://doi.org/10.5194/gmd-17-6195-2024>
- Wofsy, J., & Kuang, Z. (2012). Cloud-resolving model simulations and a simple model of an idealized Walker cell. *Journal of Climate*, 25(23), 8090–8107. <https://doi.org/10.1175/jcli-d-11-00692.1>
- Wu, M., Li, C., Collins, M., Li, H., Chen, X., Zhou, T., & Zhang, Z. (2024). Early emergence and determinants of human-induced Walker circulation weakening. *Nature Communications*, 15(1), 9161. <https://doi.org/10.1038/s41467-024-53509-6>
- Yano, J.-I., Grabowski, W. W., & Moncrieff, M. W. (2002). Mean-state convective circulations over large-scale tropical sst gradients. *Journal of the Atmospheric Sciences*, 59(9), 1578–1592. [https://doi.org/10.1175/1520-0469\(2002\)059<1578:msscol>2.0.co;2](https://doi.org/10.1175/1520-0469(2002)059<1578:msscol>2.0.co;2)
- Zhang, Y., & Fueglistaler, S. (2020). How tropical convection couples high moist static energy over land and ocean. *Geophysical Research Letters*, 47(2), e2019GL086387. <https://doi.org/10.1029/2019gl086387>

References From the Supporting Information

- Collins, W. D., Rasch, P. J., Boville, B. A., Hack, J. J., McCaa, J. R., Williamson, D. L., et al. (2006). The formulation and atmospheric simulation of the community atmosphere model version 3 (CAM3). *Journal of Climate*, 19(11), 2144–2161. <https://doi.org/10.1175/JCLI3760.1>
- Pauluis, O., & Garner, S. (2006). Sensitivity of radiative-convective equilibrium simulations to horizontal resolution. *Journal of the Atmospheric Sciences*, 63(7), 1910–1923. <https://doi.org/10.1175/jas3705.1>